James Han

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EXPERIENCE

AI Research Engineer – UIUC Siebel School of Computing and Data Science Dec 2025 – Present

- Built an end-to-end LLM agent for wearable health data, using LangChain tool calling to route queries between 100K+ wearable records stored in SQL and RAG-based scientific knowledge for sleep and HRV questions.
- Designed an agent evaluation framework with 200+ test cases benchmarking 3 LLMs across tool-use accuracy, structured-data correctness, retrieval faithfulness and relevance, and final-answer quality.
- Improved RAG faithfulness from 0.78 to 0.91 and context recall from 0.74 to 0.88 through RAGAS-driven failure analysis and tuning of chunking, metadata filtering, and retrieval parameters.
- Productionized and deployed the agent as a Dockerized FastAPI service on GCP, integrating asynchronous model and tool execution with SQL and vector-database access in a modular backend architecture.
- Added LangSmith observability across agent workflows to trace model and tool execution, identify latency bottlenecks and failures, and reduce median response time by 31% and per-query LLM cost by 24%.
- Implemented guardrails for health-related responses using groundedness checks, structured-output validation, and fallback logic, reducing unsupported or malformed answers by 37%.

NLP Research Intern – UCSB Linguistics Oct 2023 – Jan 2024

- Fine-tuned a BERT-based dialogue classification model, improving F1 score by 14.8% through hyperparameter tuning, regularization, early stopping, and class-imbalance handling.
- Built a multiprocessing NLP pipeline over ~50GB of unstructured data, automating preprocessing and embedding generation to reduce data processing time by 40%.

Data Science Intern – UCSB Ecological Restoration Apr 2023 – Jun 2023

- Developed a GCP-based thermal-image classification model to predict endangered bird egg hatching, reducing manual nest checks by 80% while achieving 98.3% precision and 100% recall.
- Built a labeled thermal-imaging dataset across 15 species in collaboration with ecology researchers, supporting model training, validation, and reliable deployment on real-world field data.

PROJECTS

Multi-Agent LLM Debate Framework | Python, Multi-Agent Systems, Prompt Engineering, LLM Evaluation Aug 2025 – Jan 2026

- Developed a multi-agent legal reasoning framework with adversarial debaters and a judge, comparing IRAC-structured and vanilla debate across 4 LLMs on up to 500 multiple-choice and 168 open-ended legal questions.
- Built a multi-metric evaluation pipeline spanning MCQ accuracy, Citation F1, BERTScore, and LLM-as-Judge, achieving 75.0% MCQ accuracy and improving Mistral 7B's normalized judge score from 0.57 to 0.82 versus a single-agent baseline.

End-to-End Churn & Retention Pipeline | PySpark, SQL, XGBoost, scikit-learn, Pandas Apr 2025 – Oct 2025

- Built an end-to-end churn risk pipeline for 96K+ customers, aggregating customer behavior and delivery features and benchmarking XGBoost, Logistic Regression, and Random Forest; XGBoost achieved 0.797 ROC-AUC / 0.967 PR-AUC.
- Extended the pipeline with RFM and review-sentiment features and productized churn scores in a Streamlit application that ranks high-risk customers, recommends retention interventions, and estimates expected savings, cost, and ROI.

EDUCATION

University of Illinois at Urbana-Champaign (**UIUC**) – *M.S. in **Computer Science*** Aug 2024 – Dec 2025

University of California, Santa Barbara (**UCSB**) – *B.S. in **Statistics & Data Science*** Sep 2020 – Mar 2023

SKILLS

Languages & Backend: Python, SQL, FastAPI, REST APIs, Git, PostgreSQL
LLM & Agents: LangChain, LangGraph, RAG, Tool Calling, MCP, LLM Evaluation, Pinecone
ML & AI: PyTorch, Transformers, scikit-learn, XGBoost, Fine-tuning (LoRA/PEFT), Hugging Face
Cloud & MLOps: Docker, GCP, AWS, LangSmith, CI/CD, Kubernetes